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MODEL SELECTION AND VALIDATION REPORT ( arctic_biomass model )
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## OBSERVATIONS:863

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( arctic_biomass model )

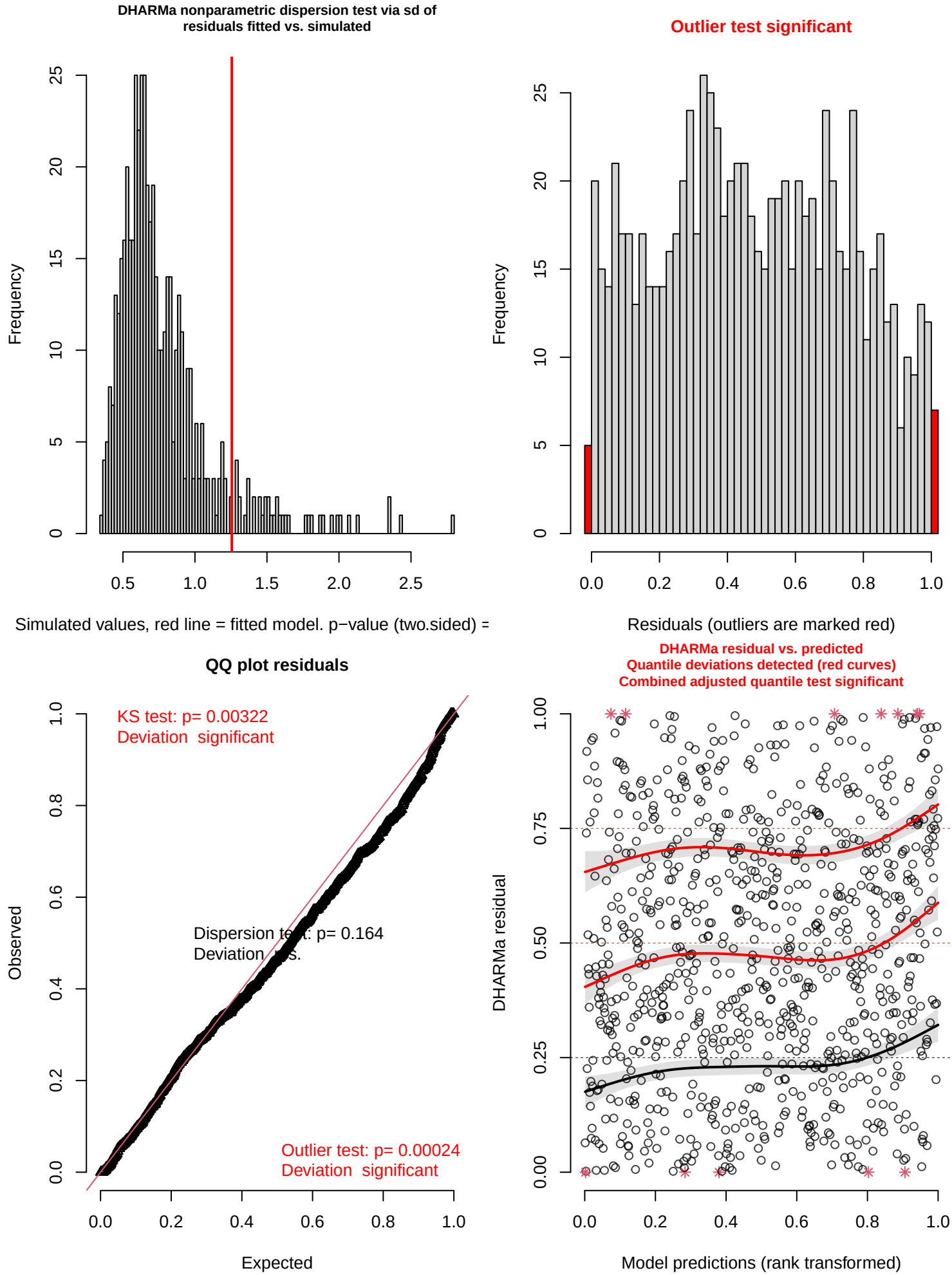
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

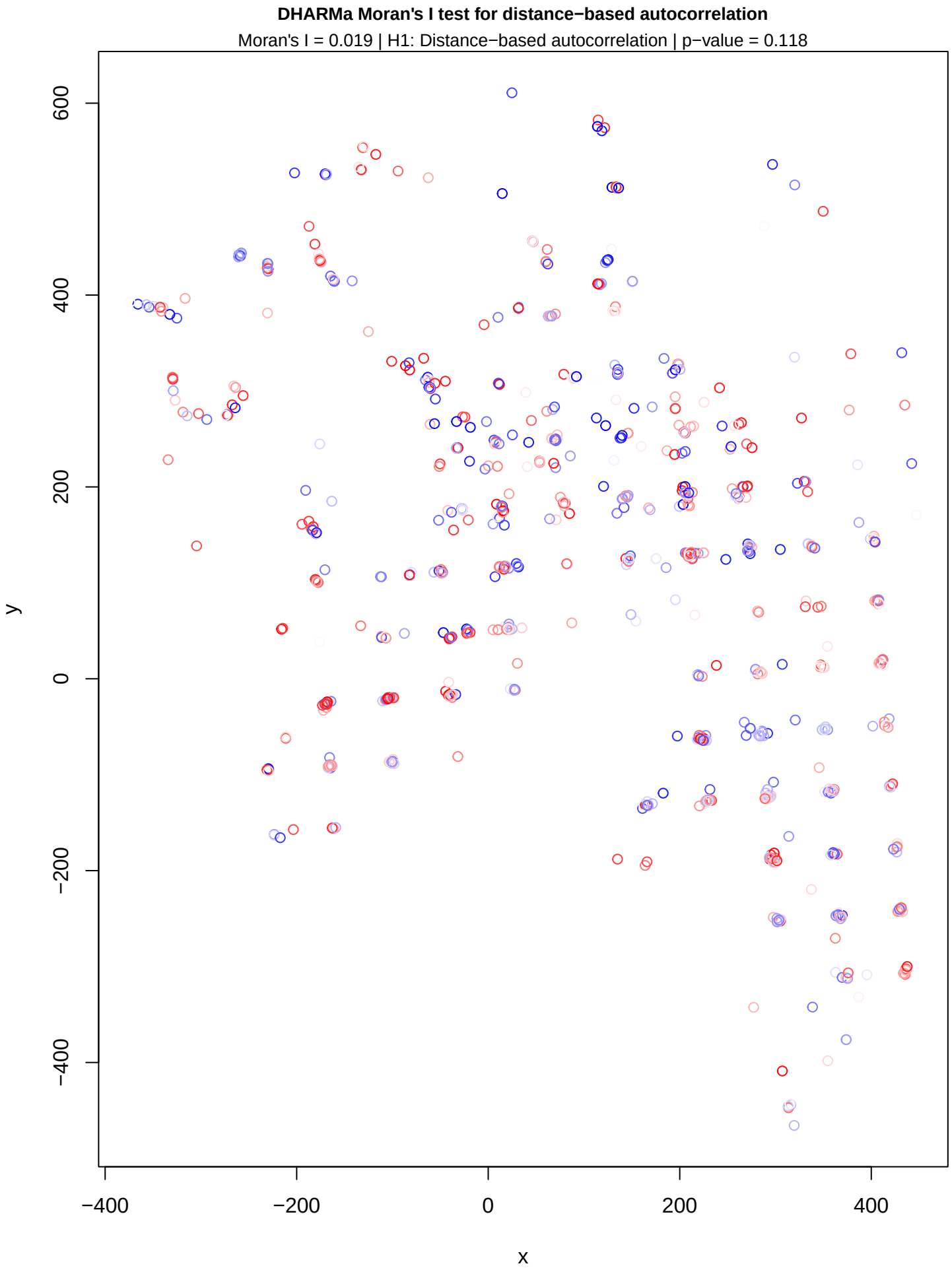
Conditional model:
              coef.est coef.se
(Intercept)       7.02    0.73
kde_fric          0.13    0.04
kde_feve         −0.15    0.04
tric             −0.14    0.04
teve             −0.32    0.04
sbt_mean          0.16    0.04
sic_mean          0.04    0.09
sbt_sd            0.16    0.07
sst_sd            0.08    0.06
log_chla_mean     0.01    0.05
depth             0.09    0.05

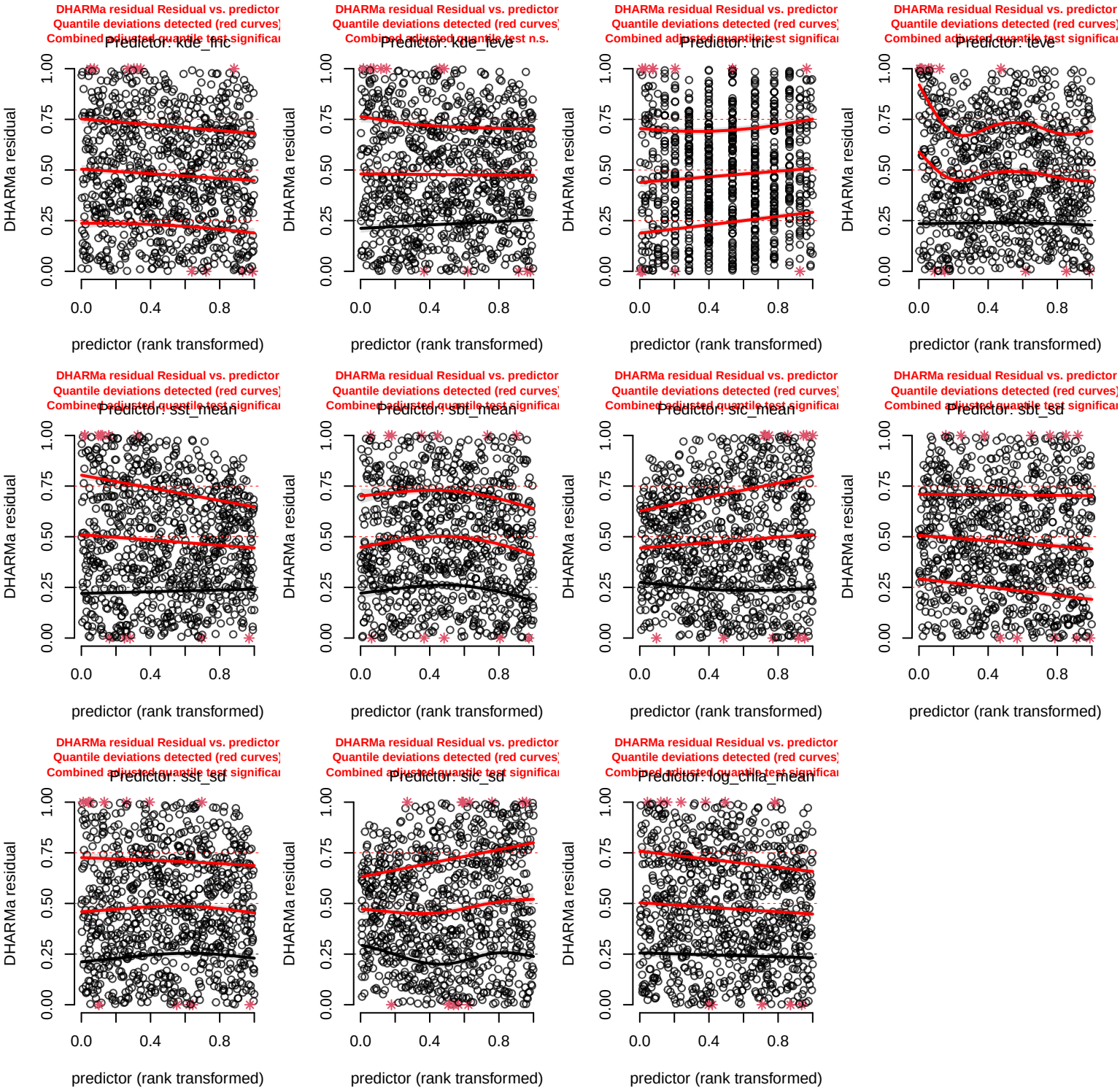
Dispersion parameter: 0.88
Spatiotemporal AR1 correlation (rho): 0.96
Matérn range: 726.22
Spatiotemporal marginal AR1 SD: 0.95
REML criterion at convergence: 7062.352

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                   df      AIC delta_AIC
spatiotemporal_random_fields_ar1                   15 14154.70  1.787931
spatiotemporal_random_fields_ar1_with_anisotropy 17 14156.49 58.471940
spatial_random_fields                              14 14214.96  3.686797
spatial_random_fields_with_anisotropy              16 14218.65  3.790045
spatiotemporal_random_fields_iid_with_anisotropy 16 14222.44  3.178125
spatiotemporal_random_fields_iid                   14 14225.62 95.221203
error_only                                         12 14320.84        NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
Simulated values, red line = fitted model. p−value (two.sided) =
Outlier test significant
Frequency
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.00322
Deviation significant
Dispersion test: p= 0.164
Deviation n.s.
Outlier test: p= 0.00024
Deviation significant
Observed
Expected
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.019 | H1: Distance−based autocorrelation | p−value = 0.118
600
400
200
0
−200
−400
y
−400
−200
0
200
400
x


DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: kde_fnc
Combined adjusted quantile test n.s.
Predictor: kde_feve
Predictor: tric
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8
predictor (rank transformed)